

1. (a) State the axiom of completeness

(b) Let A be a non-empty bounded set of reals. Prove that there is a sequence $a_n \in A$ such that $a_n \rightarrow \sup A$.

2. (a) State the definition of well-ordered principle of natural numbers, and state the definition of a countable set.
- (b) Assume that A is an infinite set. Prove that A contains a countable subset.
- (c) Let $P_n(\mathbb{Z}) = \{a_0 + a_1x + \cdots + a_nx^n, a_0, a_1, \dots, a_n \in \mathbb{Z}\}$, the set of all polynomials of power at most n with integer coefficients. Prove that $P_n(\mathbb{Z})$ is a countable set.

3. (a) State the definition of a sequence, $a_n \rightarrow a$.

(b) Using (a) show that $\lim_{n \rightarrow \infty} \frac{2n+1}{3n-5} = \frac{2}{3}$.

(c) Using (a) show that if $a_n \geq 0$ and $a_n \rightarrow a$ then $\sqrt{a_n} \rightarrow \sqrt{a}$

4. (a) State Bolzano-Weierstrass Theorem

(b) Let $0 < a_1 < 1$, and the sequence defined recursively by $a_{n+1} = a_n(2 - a_n)$. Prove by induction that $0 < a_n < 1$.

(c) Use (b) to prove that a_n is a monotone sequence, and find the value of the limit, a , $a_n \rightarrow a$.

5. (a) State the definition of a series $\sum_{n=0}^{\infty} a_n$ is convergent

Determine if the following series is convergent or divergent and using (a) to prove your answer.

(b) $\sum_{n=1}^{\infty} \frac{1}{n}$

(c) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$

6. Determine whether the following statements are true (**T**) or false (**F**); if choosing **T**, must provide with a reason or the statement of a theorem, if choosing **F**, you must provide with a counterexample.
- (a) Given sets, A and B , if $A \times B$ is countable, then both A and B are countable
 - (b) There is a set A whose cardinality is strictly greater than the cardinality of $\mathbb{R}$
 - (c) Given two sequences, a_n, b_n if $a_n b_n$ converges then either a_n or b_n converges.
 - (d) If $a_n \rightarrow a$ with a_n explicitly defined, then one is able to identify the value a .
 - (e) Given two series, $\sum a_n, \sum b_n$ if $\sum a_n^2$ and $\sum b_n^2$ converge, then $\sum a_n b_n$ converges.
 - (f) If $\sum_{n=1}^{\infty} a_n = a$, then $|a_n| \rightarrow 0$ as $n \rightarrow \infty$.